

2(a)(i)	<i>any four from:</i> <i>differences</i> 1 cotton (enzyme / rubisco activase) has a <u>higher</u>, optimum temperature / temperature at which activity is highest ; ora 2 maximum / highest / peak, activity <u>higher</u> in flax (enzyme / rubisco activase) (than in cotton) ; ora 3 cotton (enzyme / rubisco activase) can work at <u>higher</u> temperature (than flax) ; 4 (at stated temp) <30 °C flax (enzyme) has <u>higher</u> activity than cotton ; ora 5 (at stated temp) >30 °C cotton (enzyme) has <u>higher</u> activity than flax ; ora <i>similarities</i> 6 (activity of) both increases then decreases as temperature increases ; 7 (enzyme / rubisco activase) activity, same / 0.104 (au), at 30 °C ;	4
2(a)(ii)	<i>any three from:</i> 1 cotton, is adapted to / lives in areas with, high(er) temperatures ; ora 2 <i>ref. to</i> different, genes / alleles (in cotton vs. flax) ; 3 <i>ref. to</i> different primary structure of rubisco activase (in cotton vs. flax) ; 4 <i>ref. to</i> different tertiary structure of rubisco activase (in cotton vs. flax) ; 5 AVP ;	3

Question	Answer	Marks
2(b)	any three from: 1 less / no, CO₂ fixation / carboxylation of RuBP ; 2 less / no, GP, made into / converted to, TP ; 3 less / no, regeneration of RuBP ; 4 less / no, glucose / hexose, made ;	3
2(c)	any two from: 1 take, gene / (c)DNA, for <u>rubisco activase</u> ; 2 from, <u>cotton</u> and insert into, (false) flax / (named crop) plant / embryo ; 3 to maintain rubisco action at, high temperatures / >37.5 °C / up to 42.5 °C ; 4 AVP ;	2